Spontaneity, ΔG

ΔG predicts the spontaneity of reaction ; $\Delta G = \Delta H - T\Delta S$

$\Delta G < 0$: spontaneous; $\Delta G > 0$: not spontaneous

Case 1: $\Delta H < 0$, $\Delta S > 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = • at high temperature, ΔG =
Case 2: $\Delta H > 0$, $\Delta S < 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = • at high temperature, ΔG =
Case 3: $\Delta H > 0$, $\Delta S > 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = • at high temperature, ΔG = ➢ at sufficiently high temperatures, the ______________ outweighs _______________, hence ΔG is ________
Case 4: $\Delta H < 0$, $\Delta S < 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = • at high temperature, ΔG = ➢ at sufficiently high temperatures, the ______________ outweighs _______________, hence ΔG is ________

Common errors to avoid when calculating ΔG

- ➢ unit conversions: ΔH is often given in kJ mol^{-1}, ΔS often in J K^{-1} mol^{-1}.
- ➢ T must be in Kelvins.
- ➢ ΔG, ΔS and ΔH can be solved by drawing energy cycles.

Notes for $\Delta G = 0$

- ➢ $\Delta G = 0$ for phase changes at its specific phase change temperature

  e.g.: $H_2O(l) \rightarrow H_2O(g)$ at 100 °C; $\Delta G = 0$

  e.g.: $H_2O(l) \rightarrow H_2O(s)$ at 0 °C; $\Delta G = 0$

Entropy, ΔS

Entropy is a measure of the disorder in the system

$\Delta S = S_{final} - S_{initial}$

$\Delta S > 0$: ___________ in disorder in the system, ___________ in entropy

$\Delta S < 0$: ___________ in disorder in the system, ___________ in entropy

System	ΔS & explanation
1 mol of $H_2O(s)$ melts to form $H_2O(l)$	• liquid particles ________________________ • there is an increase in number of ways to arrange the particles and their energies • entropy increases, $\Delta S > 0$
temperature of a sealed vessel containing 1 mol of $N_2(g)$ is raised from 300 K to 500 K	• broadening of maxwell Boltzmann distribution curve • there is an increase in ____________________ ______________________________________ • entropy increases, $\Delta S > 0$
2 mol of $SO_2(g)$ completely reacts with 1 mol of $O_2(g)$ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	• decrease in _____________________________ • there is a decrease in ____________________ ______________________________________ • entropy ____________________

Spontaneity, ΔG

ΔG predicts the spontaneity of reaction ; $\Delta G = \Delta H - T\Delta S$

$\Delta G < 0$: spontaneous; $\Delta G > 0$: not spontaneous

Case 1: $\Delta H < 0$, $\Delta S > 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = —ve • at high temperature, ΔG = —ve
Case 2: $\Delta H > 0$, $\Delta S < 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = +ve • at high temperature, ΔG = +ve
Case 3: $\Delta H > 0$, $\Delta S > 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = +ve • at high temperature, ΔG = —ve ➢ at sufficiently high temperatures, the negative $-T\Delta S$ outweighs positive ΔH, hence ΔG is negative
Case 4: $\Delta H < 0$, $\Delta S < 0$	**$\Delta G = \Delta H - T\Delta S$** • at low temperature, ΔG = —ve • at high temperature, ΔG = +ve ➢ at sufficiently high temperatures, the positive $-T\Delta S$ outweighs negative ΔH, hence ΔG is positive

Common errors to avoid when calculating ΔG

- unit conversions: ΔH is often given in kJ mol^{-1}, ΔS often in J K^{-1} mol^{-1}.
- T must be in Kelvins.
- ΔG, ΔS and ΔH can be solved by drawing energy cycles.

Notes for $\Delta G = 0$

- $\Delta G = 0$ for phase changes at its specific phase change temperature

  e.g.: $H_2O(l) \rightarrow H_2O(g)$ at 100 °C; $\Delta G = 0$

  e.g.: $H_2O(l) \rightarrow H_2O(s)$ at 0 °C; $\Delta G = 0$

Entropy, ΔS

Entropy is a measure of the disorder in the system

$\Delta S = S_{final} - S_{initial}$

$\Delta S > 0$: increase in disorder in the system, increase in entropy

$\Delta S < 0$: decrease in disorder in the system, decrease in entropy

System	ΔS & explanation
1 mol of $H_2O(s)$ melts to form $H_2O(l)$	• liquid particles moves around more freely • there is an increase in number of ways to arrange the particles and their energies • entropy increases, $\Delta S > 0$
temperature of a sealed vessel containing 1 mol of $N_2(g)$ is raised from 300 K to 500 K	• broadening of maxwell Boltzmann distribution curve • there is an increase in number of ways to arrange the energies of the particles • entropy increases, $\Delta S > 0$
2 mol of $SO_2(g)$ completely reacts with 1 mol of $O_2(g)$ $2SO_2(g) + O_2(g) \rightarrow 2SO_3(g)$	• decrease in number of mol of gaseous particles • there is a decrease in number of ways to arrange the particles and their energies • entropy decreases, $\Delta S < 0$